

Concluding thoughts on Nominalism
April 6, 2004

1 How we might define ‘electron’

Suppose that we have the notion of a “natural” class, either as primitive (Primitive Natural Class Nominalism) or defined somehow in terms of resemblance, as in Resemblance Nominalism. How might we to define ‘electron’ and other such physically basic vocabulary?

An idea: we define ‘ x is an electron’ to mean ‘ x is a member of the natural class that *plays the electron-role*’, and we define ‘ C plays the electron-role’ by looking at the role played by the predicate ‘electron’ in the *fundamental laws of nature*.

So, for example, suppose the fundamental laws of nature say something like this:

- (1) ‘Everything is either an electron, a proton or a neutron; any two electrons repel one another; any two protons repel one another; any electron and proton attract one another; and neutrons neither repel nor attract anything else.’

Then we can define ‘electron’ thus:

- (2) To be an electron is to be a member of some natural class C such that everything is either a member of C , a proton or a neutron; any two members of C repel one another; any two protons repel one another; any member of C and any proton attract one another; and neutrons neither repel nor attract anything else.

Note that any definition of ‘electron’ along these lines will entail that certain claims which we might have thought of as *contingent laws of nature* are really *necessary truths*—for example, that any two electrons repel one another, or that any electron and proton attract one another. How bad is this?

The obvious problem with (2) is the fact that we use other basic physical predicates like ‘proton’ and ‘neutron’ to define ‘electron’. How are we to define these predicates? We can’t use the same method, since that would produce a circular system of definitions.

At this point we can use an ingenious method invented by the philosopher F.P. Ramsey (active in the 20s and 30s): to eliminate circularity in a system of definitions, we replace the defined terms with existential quantifiers over classes. This gives us something like this:

- (3) To be an electron is to be a member of some natural class C such that there are two other natural classes D and E such that everything is either a member of C , a member of D or a member of E ; any two members of C repel one another; any two members of D repel one another; any member of C and any member of D attract one another; and members of E neither repel nor attract anything else.

This still uses the predicates ‘repel’ and ‘attract’; but in principle we could ‘Ramsify’ these too, using (e.g.) the notion of a natural class of ordered pairs.

Universals
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2 Theories of universals

Some questions for a theory of universals and particulars:

- (1) Can a universal exist without being instantiated by any particular?
- (2) Can a particular exist without instantiating any universal?
- (3) Are there any *logical relations* among universals: can we ever truly say, for example, that one universal is the *conjunction*, or *disjunction*, or *negation* of another? If so, which logical relations can obtain among universals?
- (4) Are there universals corresponding to *every* predicate (where u corresponds to ' F ' if it is necessary that u is instantiated by all and only those things that are F)? If not, how might we determine whether any given predicate has a universal that corresponds to it?
- (5) Are universals *necessary* or *contingent* beings?
- (6) Are universals located in space or time? If so, where?
- (7) What are the mereological relations between universals and particulars? Are universals *consituents* of the particulars that instantiate them? Are particulars *constituents* of the universals that instantiate them?
- (8) Can there be "higher-order" universals, universals which are instantiated by other universals?

3 Armstrong's theory of states of affairs

Armstrong holds that whenever a particular x instantiates a universal u , there is a *state of affairs* with x and u as its constituents. Moreover, he holds that it is *only* in this case that there is a state of affairs with x and u as constituents. So perhaps he has in mind a *definition* of what it is for a particular to instantiate a universal, something like this:

For x to instantiate u is for it to be the case that there is a state of affairs with x as its particular constituent, and u as its universal constituent.

There are relational states of affairs as well as monadic ones: they have more than one particular constituent.

4 Thin and thick particulars

The things we've just been calling 'particulars'—the constituents of states of affairs—Armstrong calls 'thin particulars', things 'taken apart from' their properties. He also recognises a category of "thick" particulars. For any thin particular x , the thick particular corresponding to x can be taken to be the state of affairs having x as its particular constituent and the conjunction of all universals instantiated by x as its universal constituent. (An alternative definition, which Armstrong takes to be equivalent to this: the thick particular is the mereological sum (fusion) of all the states of affairs having x as their sole particular constituent.)

Thick particulars can also be said to "instantiate" universals, though not in the same sense as thin particulars: we can say that a thick particular instantiates a universal iff it has it as a constituent.

If there are these two sorts of particulars, what are the ordinary particulars of our ordinary talk, things like books and people and electrons?

Armstrong seems to suggest that they are thick particulars, that thin particulars are the sorts of things only philosophers would want to talk about.

One problem for this view: isn't it the case that whenever a thick particular "instantiates" a universal, it does so essentially, i.e. that it can't exist without "instantiating" that universal? Armstrong certainly seems to believe that states of affairs have their constituents essentially, and that conjunctive universals have their conjuncts essentially.

5 The truthmaker principle

Armstrong argues for the existence. 88): ‘for every contingent truth at least, there must be something in the world that makes it true’.

What must be the case for x to make it true that P ? At least this: it must be *necessary* that if x exists, then P . So if x could exist without being F , x itself can’t be a truthmaker for the truth that x is F . By contrast, the state of affairs that x is F , if there is such a thing, *is* an object which couldn’t exist unless x were F : so it is fitted to be the truthmaker for the truth that x is F .

Why are we supposed to believe the truthmaker principle? Armstrong doesn’t argue for it: he thinks it’s obvious (p. 89). I disagree.

Some famous hard cases for the truthmaker principle: *negative* truths (‘It’s not the case that Jones is a spy’) and *universal* truths (‘All dogs have fleas’). Any truthmakers for these truths will have to be pretty strange entities.

Is there a good argument from the truthmaker principle to the claim that there are states of affairs? To the claim that there are universals?